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A Hybrid Model for Investment Prioritization and Performance Analysis in Electrical Power Distribution Systems

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Abstract
This paper presents a methodology developed to prioritize feeders for receiving investments and analyzing a power utility distribution network. The proposal applies the Analytic Hierarchy Process method to determine which feeders should receive the available resources as a priority. The methodology provides greater assertiveness in decision-making, besides using historical data, the opinion and experience of company specialists. The Plan-Do-Check-Action model assists in verifying improvement interventions, allowing the visualization of the network's stretches with recurring failures, expanding the possibilities for improvement. The correct application of resources is fundamental to the company, as it can guarantee favorable results in feeders' performance, thus improving the company's indicators and the perception that customers have of the utility. The scientific contribution of this paper is the development of a hybrid technique that combines the prioritization method with a model to check its effectiveness, with flexibility, and versatility. In addition, the computational tool developed brought significant time savings for data extraction and analysis, allowing the user to customize the research, defining the periods, types of equipment, and causes of failure easily and quickly.

Keywords Power Utility · Feeders' Performance · Analytic Hierarchy Process · PDCA · Continuity Indicators

1 Introduction

Electricity is an increasingly valuable asset in people's lives. It is present in all branches of society, in economic activities, everyday life, and leisure. For having this vital role, the electric power distribution system must always be improved, presenting stability, reliability, safety, and quality.

Maintaining the power grid's stability and efficiency requires constant monitoring and analysis (Duque et al., 2021). In addition, system failures can cause economic losses for both consumers and utilities, which is another reason to maintain supply quality (Karimi & Jannati-Oskuee, 2021).

In Brazil, after the privatization of the electricity sector in the 1990s, ANEEL, the Brazilian Electricity Regulatory Agency, began to monitor the industry through the analysis of indicators such as continuity and distribution network loading, aiming, in this way, to guarantee the quality of the service provided by the utilities. Another significant change that occurred with privatization was decentralizing decision-making on investments in the sector (Costa et al., 2020). Bradshaw (2017) reviews the Brazilian electrical system's evolution, showing this scenario transition. This change in paradigm gave greater autonomy to the sectors giving freedom over the application of investments. Therefore, maintaining the quality of service has become the responsibility of the utilities.

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Consumers have come to demand improvements in the sector. With the universalization of access to electric energy (Pereira et al., 2010), many consumers began to access the distribution network. This increase in load reflects the need for higher quality service to supply the demand, leading consumers to pressure the regulatory agency to meet their wishes. In this context, ANEEL has been regulating the sector to obtain continuous electricity supply with quality improvements. The agency has created quality indicators to monitor the utilities' performance, applying severe fines in case of non-compliance (Soares et al., 2014).

ANEEL (2016) published Procedures for Electric Energy Distribution in the National System (PRODIST). The module 8 of PRODIST regulates the power quality factors. By controlling interruptions in energy supply and the disclosure of continuity indicators, customers and the supervisory body can verify its quality. The leading continuity indicators are the Duration of Interruptions per Consumer Unit (DEC), Equivalent Interruption Frequency Consumer Unit (FEC), equivalent to SAIDI (System Average Interruption Duration Index), and SAIFI (System Average Interruption Frequency Index) indices.

The distribution network needs constant investments in maintenance and improvement to keep the operation in stable conditions. However, the financial resources available are limited, so using them assertively can be a differential for the utility's results. In this scenario, this paper proposes:

- A prioritization method capable of analyzing the utility's indicators of interest, indicating the best option for resource allocation;
- A methodology for diagnosing the distribution network's performance, allowing the analyses of quality indicators;
- A model that verifies the effectiveness of the network's interventions, highlighting failure recurrence.

The problem of prioritizing investments in distribution networks involves many issues, such as analyzing historical data, the utility's strategy, quality indicators, regulations, and others. For this reason, this work approaches the use of the Analytic Hierarchy Process (AHP) method. This method's application is frequent in complex decision problems, which involve many criteria and user preferences (Colak et al., 2020).

Mussoi and Teive (2021) present a robust model for prioritizing investments, basing the analysis on a portfolio of projects. They apply the Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) and Simple Multi-Attribute Rating Technique (SMART). Considering the decision-maker profile embedded into the model by the Rank-Order Centroid (ROC) and AHP, in the end, the authors conclude that the AHP presents the same solution as other models, showing that this method is adequate.

Li et al. (2021) apply the AHP for a different purpose, but with a positive result, analyzing failures in wind turbines. The authors use experts' opinions from other areas, showing that although there are discrepancies between the weights given by the experts, the result of the method converges, so does the subjectivity of the weights. Thus, the subjectivity of applying the weights, if correctly used, did not interfere with the assertiveness of the method.

Amohadi and Firuzabad (2019) apply the AHP to define the optimal location of reclosers and disconnecting switches. The study's promising results show that the method is robust when used with other techniques in power distribution networks. Wu et al. (2019) apply the AHP to analyze voltage stability. They combine the AHP with the entropy method to define the weights for ten voltage stability criteria, identifying the most stable and the most unstable bus in the distribution network.

Just applying a prioritization method may not be enough to achieve more quality in the electric power supply. The Plan-Do-Check-Act (PDCA) cycle is a tool that assists in the control and execution of improvement activities (Song & Fischer, 2020). This methodology acting in conjunction with the prioritization method can create a precise continuous improvement process.

In this study, the reports generated by the utility have a table format. Therefore, this work developed a tool with Microsoft Excel and Visual Basic for Application (VBA) to generate the analyses. Other authors use VBA for analysis and learning tools such as Suwanasri et al. (2020) and Demir et al. (2018). The data come from the financial sector, and the reports of failures in the distribution network. These data show, among other information, the time, equipment (location), duration, number of customers affected, and the cause of the failure occurrences.

This study was conducted and applied in a real distribution system. The Regional Energy Cooperative Taquari Jacuí, CERTAJA Energia, distributes electricity to more than 28,000 cooperative members and customers, has approximately 4000 km of networks located in 19 municipalities of Rio Grande do Sul—Brazil.

Prioritization of maintenance projects generates strategic research. Pierre and Arguello (2018) present a model for prioritizing resources based on system reliability, using SAIDI and SAIFI indicators along with the cost and impact of interventions. The authors apply a probability density algorithm to make interrupt predictions. Creating predictions for one or more years, the authors base prioritizations on these future scenarios. The proposal suggests gains in reliability in different simulated scenarios. Wu et al. (2019) propose a model that analyzes a portfolio of projects to define those with greater urgency. The authors analyze customer needs, the economic benefit of the works, network reliability, and energy loss. Both works (Pierre & Arguello, 2018; Wu et al. 2019) present

promising results regarding the prioritization of investments, but they do not point to a metric to monitor the projects after execution.

Afzali et al. (2019) present a prioritization approach based on the most critical equipment, considering the cost that equipment failure can cause to the customer. The authors develop a variable reliability index that depends on the execution of maintenance actions. It is not possible to observe the gain that the works will bring in the continuity indicators as this proposal presents.

Carnero and Gómez (2017) present a model for choosing the best maintenance strategy based on the local context of the equipment, with a focus on hospital areas. Chu et al. (2009) consider reliability indices such as SAIDI and SAIFI to define maintenance priorities. The authors create a customer failure impact index to determine the importance and weigh choices. Dehghanian et al. (2012) use AHP combined with Fuzzy logic. The authors define the critical equipment to receive investments. The method uses qualitative and quantitative aspects in the analysis.

All works present promising results regarding the prioritization but do not present all the features proposed here. In this paper, we offer an investment prioritization model, which can be proven effective in the intervention analysis module. In addition to presenting the reliability indicators quickly to the operator, we show the possible gain that the prioritized intervention will bring for each reliability indicator.

This paper presents the following structure. Section 2 presents the prioritization method. Section 3 approaches the developed network performance analysis. Section 4 discusses the effectiveness of investments made to improve the distribution network, and finally, Sect. 5 concludes the work.

2 Prioritization Method

The DEC and FEC continuity indicators, the number of sustained interruptions in the energy supply, the number of customers, and the Distribution System Usage Rate (EUSD) are indicators considered a priority by the utility company. They will be part of the prioritization process used in this study.

As mentioned before, DEC and FEC indicators are equivalent to SAIDI and SAIFI. EUSD is the charge for the use of distribution facilities. It is obtained by the product of the use tariffs (R$—Brazilian Real), by the respective contracted or verified amounts of distribution system use (kW) and energy (kWh). It is calculated monthly per consumer unit. To obtain the total EUSD value of a switch, feeder, or even a period, add the value of each consumer unit involved. Regarding the sustained outages, there is a greater probability of assertiveness in identifying the cause of the failure because a technical team analyzed and resolved the problem. For this reason, they are considered.

Table 1 AHP preference scale

Numeric scale	Verbal	Scale definition
1	Equal	Elements of equal preference
3	Moderate	Slightly more preferred element
5	Strong	Strong preference
7	Very strong	Element much favored over the other
9	Extremely strong	Element preference is clear and indisputable
2, 4, 6		Intermediate levels

The AHP method evaluates alternatives in a multicriteria decision scenario, which fits this case study well. The model performs pairwise comparisons based on expert opinion. The comparisons used a judgment scale to determine how dominant one element is over another (Saaty, 2008). According to Nassiri Pirbazari and Jalilian (2020), the method is applicable when decision-making involves many options and criteria, whether quantitative or qualitative. The AHP consists of structuring the decision problem into different hierarchical levels, such as the main objective, criteria, and alternatives, followed by a pairwise comparison at each level (Mastrocinque et al., 2020). In this sense, the method needs three steps: definition of the hierarchy level, creation of the judgment matrix, and sensitivity analysis. The first hierarchical level is the objective of prioritization, in this case, the feeder of the distribution network that should receive investments. The second level contains the analyzed criteria, which are the indicators of interest. Finally, there are the alternatives, which are all the feeders of the network.

One of the method's central points is the definition of the degree of relevance of the criteria to the main objective. This step is the critical point of prioritization, and it should count on the most significant number of specialists. An error in this step can result in prioritization with a low level of certainty. Specialists from the engineering, maintenance, and operation sectors participated in the attribution of weights. Defining each criterion's relevance in decision-making means assigning a metric for each item that is, giving a value so that all components can be related to each other. For this step, the reference is Table 1 shown at Saaty (2008).

The quantification of the criteria happens in pairs. Table 2 summarizes this step, showing which element is more relevant for the utility's strategy. The symbol "+" shows which item has the preference over the pair. The criterion EUSD is the most pertinent since it has a higher weight than all others. The second most impacting is the DEC.

The following steps of the method organize the data into a matrix of pairs and calculate the matrix's eigenvalues and

Table 2 Assigning the weights

Criterion 1		Weight		Criterion 2
DEC impact	+	5	–	FEC impact
DEC impact	+	2	–	Number of occurrences
DEC impact	+	2	–	Number of consumers
DEC impact	–	3	+	EUSD
FEC impact	–	5	+	Number of occurrences
FEC impact	–	5	+	Number of consumers
FEC impact	–	9	+	EUSD
Number of occurrences	–	1	+	Number of consumers
Number of occurrences	–	5	+	EUSD
Number of consumers	–	5	+	EUSD

Table 3 Preference matrix

	DEC	FEC	Occur	Cons	EUSD	EV	Pref
DEC	1	5	2	2	0.3	1.8	0.21
FEC	0.2	1	0.2	0.9	0.1	0.2	0.03
Occur	0.5	5	1	1	0.2	0.8	0.12
Cons	0.5	5	1	1	0.2	0.8	0.12
EUSD	3	9	5	5	1	3.6	0.52
Sum	5.2	25	9.2	9.2	1.84	7.1	1

Table 4 Priority setting—DEC

Feeder	DEC (h)	Normalization (%)
FD 1	0.631	2.333
FD 2	0.711	2.627
FD 3	3.082	11.390
FD 4	1.322	4.886
FD 5	1.389	5.132
FD 6	2.581	9.540
FD 7	3.843	14.201
FD 8	4.388	16.216
FD 9	2.093	7.735
FD 10	1.674	6.185
FD 11	1.781	6.581
FD 12	0.692	2.556
FD 13	2.873	10.618

Table 5 Example of a decision matrix

Criteria	DEC	FEC	Occur	Cons	EUSD	Result
Weights	0.20	0.03	0.12	0.12	0.51	–
FD 1	0.02	0.04	0.03	0.01	0.02	2.2%

eigenvectors, determining the criteria' ranking. Finally, it is necessary to calculate the consistency ratio, which indicates the level of coherence of the weights assigned by the experts. Saaty (2008) presented the details of the calculations used in the method.

Table 3 shows the preference matrix with the calculated data. These data yield a consistency ratio of 0.028, which makes the analysis adequate. The main diagonal has values equal to 1 because there is the intersection of the same criteria. The values below the diagonal are the inverse of the values above due to the weights assigned by pairs. For example, if the DEC weights 5 compared to FEC, the FEC will weigh 0.2 in relation to the DEC.

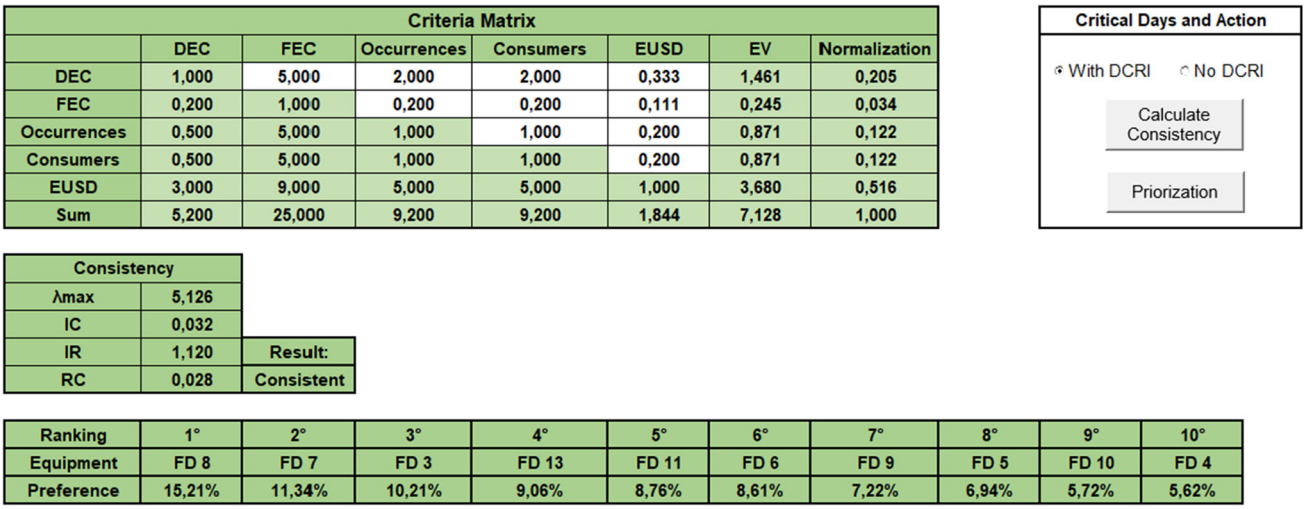
The first step of the method defines the criteria and how each one influences the decision-making process. The next stage is to determine the priorities; this definition is necessary to calculate each criterion's relevance within a given period. All failure data that occurred in 2017 are part of this study, and each alternative's priority depends on its result in the period. For example, for the DEC criterion, each feeder's priority is the percentage representation of the feeder's DEC about the utility's total DEC. This analysis extends to all criteria.

Table 4 shows an example of this for the DEC. The first column shows the fictitious feeder's name, created to keep the utility's data confidential. The indicator's actual value is in the middle column, while the third column shows the normalized value in percentage form. These data allow to prioritize the equipment through the decision matrix, multiplying the calculated priorities, as presented in the previous tables, by the weight defined by the experts, shown in Table 4.

Table 5 presents an example the data from feeder FD 1. The result column shows the sum of the previous columns' products, this value represents the percentage value of the feeder preference.

Figure 1 shows the tool screen, with the feeders' ranking according to the method, the criteria matrix with the weights defined by the experts, and the consistency check. The feeder FD 8 obtained 15.21% of preference, prioritizing for analysis for having the highest preference rating among all the equipment. The tool allows the option of using critical days using a flag. The days receive this classification when the occurring failures exceed the period's average plus five standard deviations.

All tables presented are part of the calculations necessary for the application of the AHP method. However, they are not part of the interface of the tool. In the screen shown in Fig. 1, the user can prioritize or change each criterion's weights by recalculating its consistency ratio, giving flexibility to the tool, allowing it to follow the utility's strategy, which may vary over time. For example, the decision-maker can quickly change all the criteria' weights and perform the prioritization again, choosing which scenario will benefit the utility's strategy.



Criteria Matrix

	DEC	FEC	Occurrences	Consumers	EUSD	EV	Normalization
DEC	1,000	5,000	2,000	2,000	0,333	1,461	0,205
FEC	0,200	1,000	0,200	0,200	0,111	0,245	0,034
Occurrences	0,500	5,000	1,000	1,000	0,200	0,871	0,122
Consumers	0,500	5,000	1,000	1,000	0,200	0,871	0,122
EUSD	3,000	9,000	5,000	5,000	1,000	3,680	0,516
Sum	5,200	25,000	9,200	9,200	1,844	7,128	1,000

Consistency		
λmax	5,126	
IC	0,032	
IR	1,120	Result:
RC	0,028	Consistent

Ranking	1°	2°	3°	4°	5°	6°	7°	8°	9°	10°
Equipment	FD 8	FD 7	FD 3	FD 13	FD 11	FD 6	FD 9	FD 5	FD 10	FD 4
Preference	15,21%	11,34%	10,21%	9,06%	8,76%	8,61%	7,22%	6,94%	5,72%	5,62%

Fig. 1 Applying the AHP method

3 Performance Analysis

Besides applying the AHP method, the tool developed in this study allows other possibilities. Such as seasonality analysis, verification of failure causes, the opening of feeders in smaller network sections, as, for example, areas protected by knife switches or fuse switches, and a network analysis method based on a scatter plot.

3.1 Quadrant Analysis

This method of graphical analysis separates the indicators used into two groups: performance and criticality. Performance is the sum of the continuity indicators, DEC and FEC. The criticality shows the potential for fines that the equipment has, calculated by adding the EUSD and the number of customers.

A scatter plot separated into four quadrants facilitates visualization. The further to the right, the equipment's position is, the higher its criticality—similarly, the more at the top, the worse the performance of the equipment. Figure 2 shows FD 8 in a more prominent position, reinforcing the indication of the AHP method. The user can change the weights of each criterion in real time, according to the utility's vision.

3.2 Fault Analysis and Network Sections

The prioritization of the feeder allows determining which actions are most appropriate to solve the equipment's problems. The system developed allows the opening of all fault occurrences in the period, presenting the Pareto chart data.

Figure 3 points out how each cause of failure contributed to the feeder's DEC. It shows the result in hours and as a percentage. The tool enables the analysis of the FEC and the number of occurrences, being necessary only to make the change in the selection box.

The four main failures of this feeder represent 87.67% of the DEC of the equipment. With this information, experts can plan specific actions to address the leading causes of failures, such as implementing a pruning plan, installing lightning rods and reclosers, getting results that are more expressive in the quality indicators performance-critical point of the feeder. The tool also allows the user to open the causes of failures by FEC and the number of occurrences, further basing the analysis.

The user can perform analysis on smaller network sections in the same interface screen as the tool developed, exploring the feeder by the protection switch closest to the fault that occurred. Figure 4 shows the twelve switches that most impacted the FEC of the studied equipment. Although that are more than 180 switches on this feeder, the top twelve are responsible for 44.74% of the FEC. This analysis allows the planning of minor works and quick results. The software also allows navigation through all the feeder's equipment using the arrows in the table's upper right corner.

Knowing the feeder's main problems, the user can redefine the search, selecting the switches of interest in the initial customization. After this, the fault analysis procedure will be specific for each section, making it possible to plan focused works for each section of the network or even applying the AHP method again, this time prioritizing the key that should receive the investments.

3.3 Seasonality Analysis

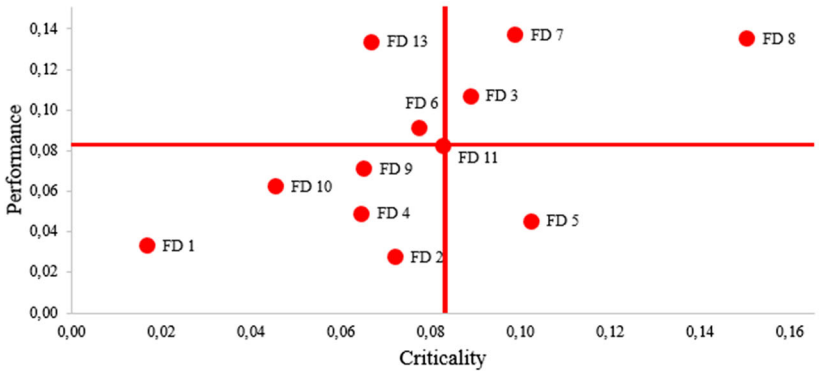
Another analysis that the tool allows is the visualization of an equipment's performance over time, with monthly openings.

Weights	
DEC	0,5
FEC	0,5

Weights	
Consumers	0,5
EUSD	0,5

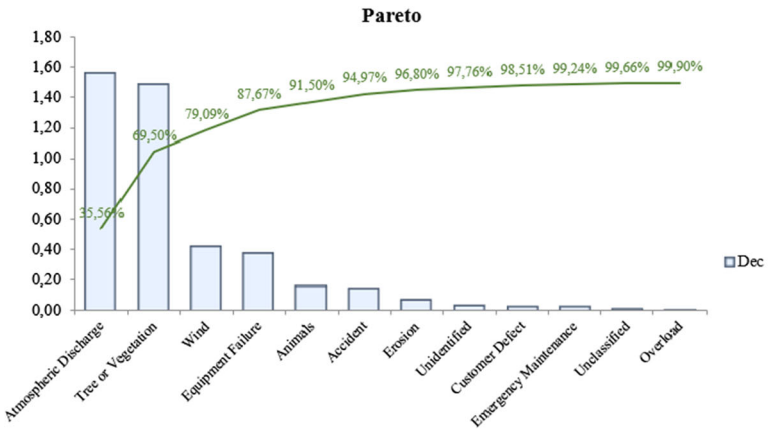
Critical Days: With DCRI / No DCRI

Data: Search Data



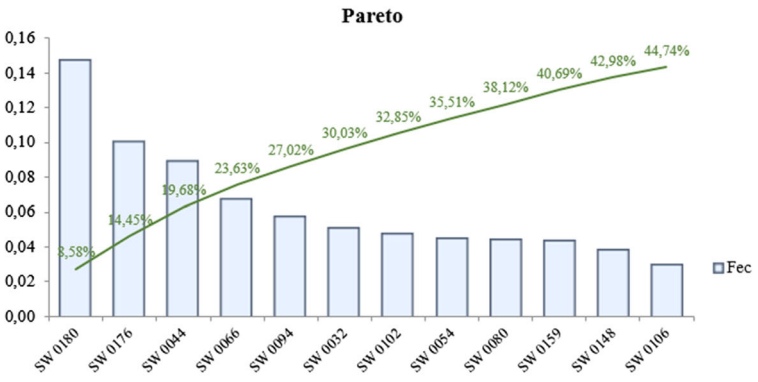
Prioritization by Quadrants		
Equipments	Performance	Criticality
FD 7	0,137	0,099
FD 8	0,136	0,150
FD 13	0,134	0,067
FD 3	0,107	0,089
FD 6	0,092	0,077
FD 11	0,083	0,083
FD 9	0,071	0,065
FD 10	0,063	0,045
FD 4	0,049	0,065
FD 5	0,046	0,102
FD 1	0,034	0,017
FD 2	0,028	0,072
FD 12	0,021	0,069

Fig. 2 Scatterplot—quadrant analysis



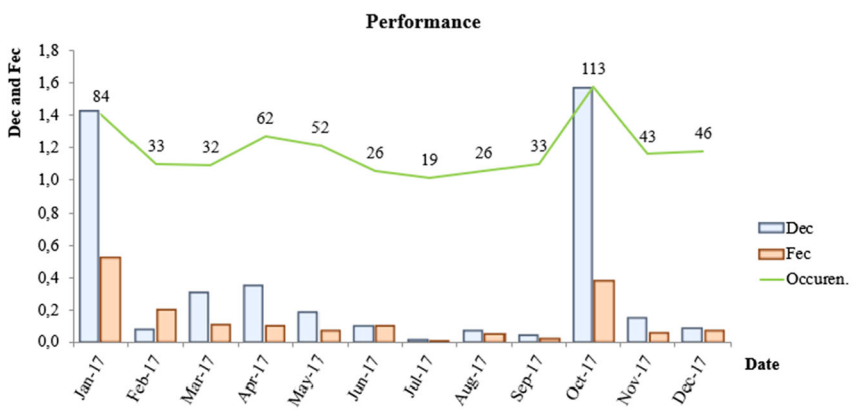
Performance by Equipment	Page 1 de 2	
Causes	Dec	%
Atmospheric Discharge	1,560233845	35,56%
Tree or Vegetation	1,489105706	69,50%
Wind	0,420923901	79,09%
Equipment Failure	0,376267648	87,67%
Animals	0,168357827	91,50%
Accident	0,15232032	94,97%
Erosion	0,07989976	96,80%
Unidentified	0,042190292	97,76%
Customer Defect	0,033250578	98,51%
Emergency Maintenance	0,031987819	99,24%
Unclassified	0,01821348	99,66%
Overload	0,010669155	99,90%

Fig. 3 Failure causes analysis



Performance by Equipment	Page 1 de 15	
Equipments	Fec	%
SW 0180	0,147422828	8,58%
SW 0176	0,100864267	14,45%
SW 0044	0,089803605	19,68%
SW 0066	0,067852158	23,63%
SW 0094	0,058120492	27,02%
SW 0032	0,051764609	30,03%
SW 0102	0,048434342	32,85%
SW 0054	0,045733746	35,51%
SW 0080	0,044812229	38,12%
SW 0159	0,044160372	40,69%
SW 0148	0,039315963	42,98%
SW 0106	0,030275683	44,74%

Fig. 4 Expanding the feeder on protection switches



Performance			Page 1 de 1	
Date	Dec	Fec	Occuren.	EUSD
Jan-17	1,426	0,526	84	R$ 309.561,4
Feb-17	0,079	0,201	33	R$ 318.892,1
Mar-17	0,312	0,108	32	R$ 356.222,7
Apr-17	0,349	0,103	62	R$ 373.604,6
May-17	0,186	0,074	52	R$ 381.923,2
Jun-17	0,100	0,101	26	R$ 350.278,0
Jul-17	0,013	0,009	19	R$ 314.220,2
Aug-17	0,072	0,054	26	R$ 298.765,6
Sep-17	0,041	0,022	33	R$ 306.575,3
Oct-17	1,571	0,380	113	R$ 297.185,5
Nov-17	0,149	0,057	43	R$ 296.162,1
Dec-17	0,090	0,076	46	R$ 295.058,8

Fig. 5 Seasonality analysis

The approach described used data from 2017, so the user can track the equipment's performance over the twelve months, as shown in Fig. 5.

This analysis allows the user to see if there have been atypical events that may be masking the feeder's performance, such as a high incidence of thunderstorms in a given month. In addition, this analysis can verify the effectiveness of interventions made in the network. The user can analyze the months before the improvement activities and the months afterward to confirm if there was a significant reduction in the service quality indicators.

4 Efficacy

A continuous improvement process should verify the effectiveness of the interventions made and plan new actions (Silva et al., 2017). In this sense, according to Prashar (2017), the PDCA method can help, through the following 4 steps: Planning (prepare and strategize); Doing (implement and operationalize); Checking (monitor and evaluate); and Acting (ensure continual improvement). Also, PDCA provides greater consistency between planning and the actual project (Nsafon et al., 2020).

Problem identification and prioritization are part of the "Planning". The application of the AHP method together with gap analysis contemplates this step. After the execution of the planned works, checking if they solved the problems identified during planning will be the next step of the PDCA cycle: the "Check". A gap analysis capable of verifying the network's performance before and after the execution of the works can realize the effectiveness of the interventions.

The first verification that the tool offers presents the DEC, FEC, and a few occurrences variation. The timeline can vary between four, eight, or twelve months. The user can define the causes of failures according to the intervention performed in the network. Figure 6 presents an example of an application, using as a database a pruning plan and taking as reference four months. The analysis is also available in a chart.

The comparison can include the months immediately before and after the actions, or the same months from the previous year, using the "Seasonal" flag. For example, suppose the activities are taken impact network performance only in the months of most significant demand, such as summer. In that case, this option needs to be enabled to highlight the result. As seen in Fig. 6, even after the activity plan's execution, failures occurred again. The PDCA cycle suggests acting after feedback. Identifying the sections with recurrent failures helps in this stage.

Actions such as a works inspection plan can provide a good response. Another option is to verify that the technician correctly identified the failure at the time of network repair. This action can mitigate the error in the new planning and prioritization cycle since the failure cause indicated will be more assertive.

Figure 7 presents data pertinent to these questions. It shows the number of sections with recurrent faults, expanding them by ranges. A list of the equipment that makes up the graph is also available via the list button.

5 Conclusion

The developed software allows visualizing the quality indicators for any medium voltage network equipment, giving the specialists one more tool to reinforce the decision-making process and even present the results obtained in a particular stretch.

The AHP method's application enables the decision-maker to choose the most assertive path for resource application since it is a robust and well-explored tool for this purpose. The process developed allows the consideration of all indicators of interest to the utility, making the attribution of weights more flexible giving the tool versatility. The application of

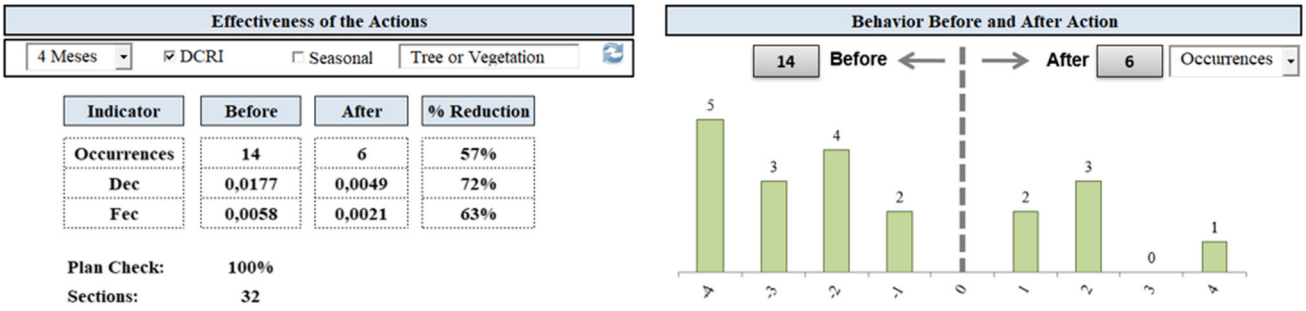


Fig. 6 Activities effectiveness—pruning plan

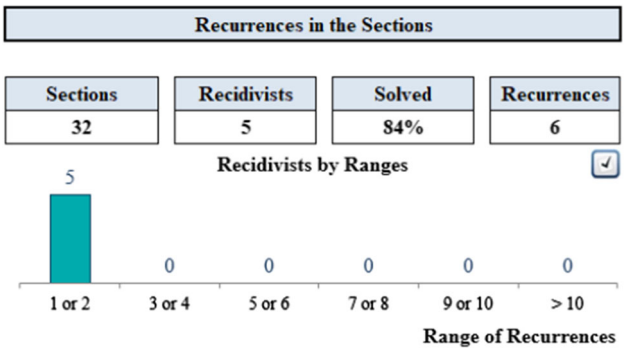


Fig. 7 Failure recurrences

the AHP method and the other analysis of failure causes, and performance allows the equipment's prioritization and even the type of work to be done.

The PDCA model complements the prioritization model since it verifies its effectiveness. Implementing this hybrid model puts the utility in a constant improvement cycle, bringing many benefits to power quality. When completing the PDCA cycle, the decision-maker can check the sections in which the failures were recurrent and investigate their reason. The plan's poor execution and the faults' seasonality are possible reasons. For example, vegetation already grown after pruning can cause a recurrence.

Still in the stage of responding to an occurrence, the technician may have mistakenly identified the root cause, which would result in the wrong planning of works causing the recurrence. This possibility of identifying problems reinforces the robustness of the method. The AHP-PDCA hybrid method offers the decision-maker a range of options to prioritize investments, analyze the results and make projections to a new stage, creating a continuous process of analysis and actions based on accurate data. Future research may explore the application of prediction techniques to estimate failure causes based on location, season, and weather conditions. This prediction can help the technician correctly indicate the root cause of the interruption in electricity supply, giving greater assertiveness in planning improvement actions.

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